

Random variable

A random variable is a collection of possible outcomes that can result from a random experiment.

Classified into two main types:

- discrete random variables
- continuous random variables

Probability Distribution

A probability distribution describes all possible outcomes of a random variable, with each outcome assigned its own probability value.

“A probability distribution shows how the values of a random variable are spread out, by assigning a probability (or probability density) to each possible value.”

Problem with Distribution

Sometimes, outcomes are so numerous or even infinite that making a table for each one becomes impossible.

Example - Height of people, Rolling 10 dice together

Solution - Function?

We can instead use a mathematical function to describe the relationship between each outcome and its probability, making analysis much easier.

$Y=f(x)$ → This function is said to be PDF

If a function $f(x)$ satisfies the conditions of a distribution function, then it is called a Probability distribution Function (PDF).

Why are Probability Distributions important?

- They help us understand the shape or pattern in our data.
- If our data matches a well-known distribution, we automatically learn a lot about its properties and behaviour.

Note: → Probability distribution and probability distribution function are often used to mean the same thing

Normal distribution

Why mean = median = mode is NOT enough?

Because a distribution can be **symmetric** (mean = median = mode) but **NOT bell-shaped**, and therefore NOT normal.

Example of symmetric but NOT normal:

- A **uniform distribution** (flat shape)

These all have **mean = median = mode**, but they are NOT normal.

When can we say a distribution is normal?

You must check **shape**, not just the center:

Methods to check normality:

1. **Histogram / density plot** (bell curve?)
2. **Q-Q plot** (points lie along straight line?)
3. **Skewness and kurtosis** (close to 0 and 3)
4. **Statistical tests** (Shapiro-Wilk, Kolmogorov-Smirnov)

Standard Normal Variate (Z)

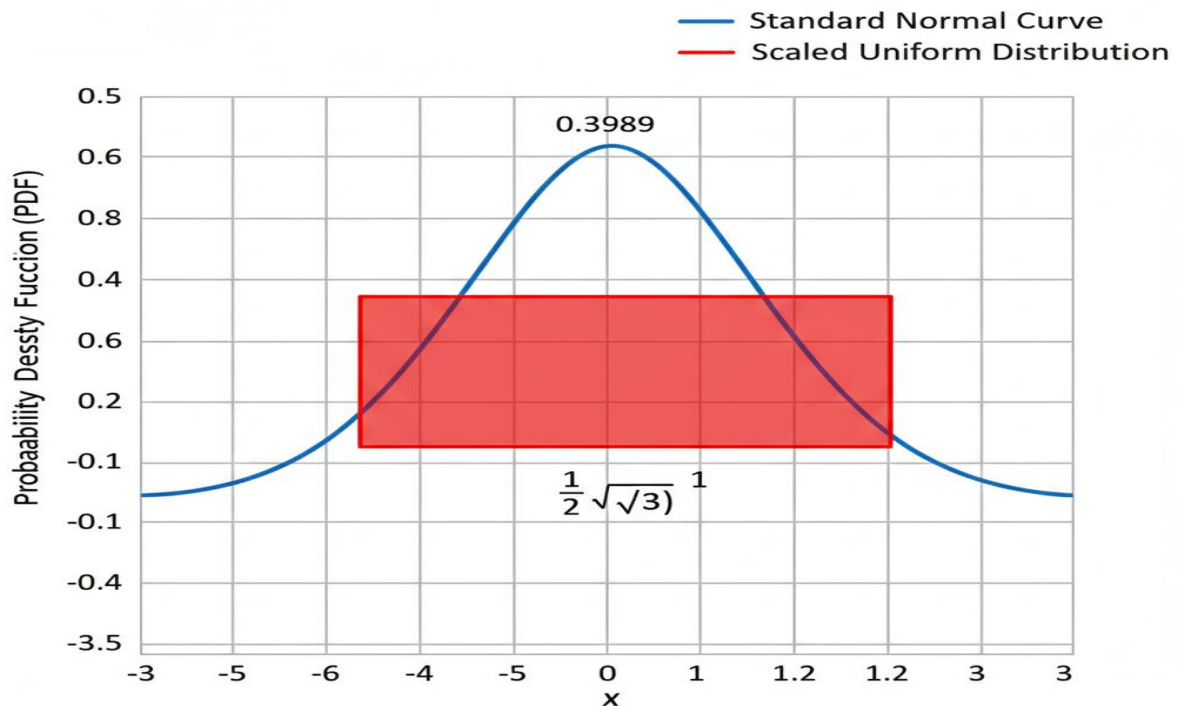
Condition 1: It **MUST** be normal

Even if mean = 0 and std = 1,
if the shape is not normal → **it is NOT a standard normal distribution.**

Example:

- A uniform distribution scaled to mean 0 and std 1
→ still **not** standard normal.

Comparison of Scaled Uniform & Standard Normal Distributions



Kurtosis

Kurtosis tells you how “heavy” or “light” the tails of a distribution are.

Kurtosis is the **fourth statistical moment** of a probability distribution.

In probability and statistics, it measures the **tailedness** of the distribution of a real-valued random

It basically answers:

How likely are extreme values (outliers)?

- **High kurtosis** → heavy tails → more extreme values
- **Low kurtosis** → light tails → fewer extreme values
- **Normal kurtosis** → normal amount of extreme values

For a **normal distribution**, kurtosis = 3.

So people often talk about **excess kurtosis** = kurtosis – 3.

Log-normal Distribution

A distribution is **log-normal ONLY IF**:

$\log(X)$ is normally distributed

A **right-skewed distribution** can be called **log-normal ONLY IF**:

$\log(X)$ is normally distributed

A random variable X is log-normally distributed if:

$$\ln(X) \sim N(\mu, \sigma^2)$$

Final Simple Rule

Log-normal $\Rightarrow$ always right-skewed

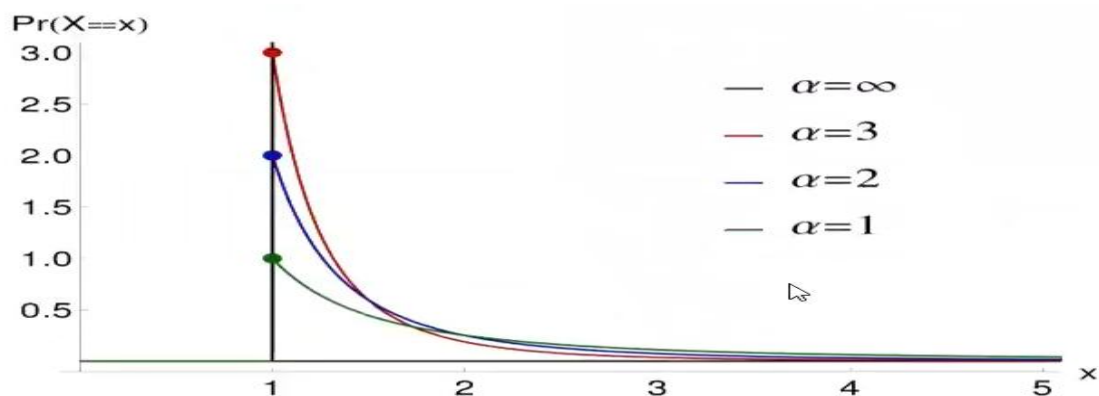
(but right-skewed does NOT always $\Rightarrow$ log-normal)

Justification #1: There are many right-skewed distributions that are **NOT** log-normal **Examples:**

- **Pareto distribution**
- **Chi-square distribution**

Pareto distribution

A **power-law probability distribution** often used to model phenomena where a small portion accounts for a large share of the total—like wealth, income, city populations, or file sizes. This is the basis of the famous **80-20 rule** (Pareto Principle): roughly 20% of causes produce 80% of effects



Mathematical Transformation

1. Function Transformer

A flexible wrapper that lets you apply any mathematical function to your data.

Transform	Formula	Use Case
Log	$\log(x)$	Right-skewed data
Reciprocal	$1/x$	Fixing heteroscedasticity
Square/Sqrt	x^2 or $\sqrt{x}$	Moderate skewness
Custom	Any user-defined function	Special cases

2. Power Transformer

Automatically finds the optimal power (λ) to make data Gaussian-like.

- **Box-Cox:** Works only for strictly positive data.
- **Yeo-Johnson:** Handles both positive and negative values, making it more flexible.

3. Quantile Transformer

This is a non-parametric, very powerful transformation.

Effect:

- Removes **all skewness**.
- Makes distribution perfectly shaped (normal or uniform).
- Great for handling **outliers** (it compresses extreme values).

But the tradeoff:

It **destroys linear relationships** (because it uses rank-based mapping).

Binomial distribution

CLT is used because it makes probability calculations simple and fast when the binomial distribution becomes complicated for large sample sizes.

Why not always use the binomial distribution?

You are **absolutely right**:

- ✓ The problem is *exactly* binomial, and
- ✓ We *can* use the binomial formula.

But look at what that involves.

Question → A researcher is investigating whether a new medication improves patient outcomes. The medication has a success rate of 75%. If the researcher enrolls 50 patients in the study, what is the probability that fewer than 35 will have a positive outcome?

$n=50$ $p=0.75$ $k = 35 - 1$

Binomial formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

To find:

$$P(X < 35)$$

You would need to compute:

$$P(X = 0) + P(X = 1) + \cdots + P(X = 34)$$

That is:

- 35 separate calculations
 - Each involving large combinations
 - Very time-consuming and error-prone
- ✗ Not practical for large n like 50, 100, 500, etc.

What CLT (normal approximation) does instead

CLT tells us:

For large n , a binomial distribution behaves like a normal distribution.

So instead of 35 calculations, we do:

1. Compute mean:

$$\mu = np$$

2. Compute standard deviation:

$$\sigma = \sqrt{npq}$$

3. Convert to a Z-score

4. Read one value from the Z-table

1. Identify the distribution

$$p=0.75$$

$$q=1-p=0.25$$

$$X \sim \text{Binomial}(n=50, p=0.75)$$

2. Check if CLT / normal approximation is valid

Rule of thumb:

$$np \geq 5 \text{ and } nq \geq 5$$

Here:

- $np = 50 \times 0.75 = 37.5$
- $nq = 50 \times 0.25 = 12.5$

✓ Both are $\geq 5 \rightarrow$ Normal approximation is valid

3. Mean and standard deviation

For a binomial distribution:

$$\begin{aligned}\mu &= np = 37.5 \\ \sigma &= \sqrt{npq} = \sqrt{50 \times 0.75 \times 0.25} = \sqrt{9.375} \approx 3.06\end{aligned}$$

4. Apply continuity correction

We want:

$$P(X < 35)$$

Using continuity correction:

$$P(X < 35) \approx P(X \leq 34.5)$$

5. Convert to Z-score

$$Z = \frac{34.5 - 37.5}{3.06} = \frac{-3}{3.06} \approx -0.98$$

6. Find the probability from the Z-table

$$P(Z < -0.98) \approx 0.1635$$

✓ Final Answer

$P(\text{fewer than 35 positive outcomes}) \approx 0.16$
--

There is about a 16% chance that fewer than 35 patients will have a positive outcome”

If the same study with 50 patients is conducted many times under the same conditions (success rate = 75%), then in about 16 out of every 100 such studies, fewer than 35 patients will show a positive outcome.

Confidence Interval

A confidence interval is a range of values, calculated from sample data, within which we expect the true population parameter (such as the population mean or proportion) to lie.

It is used to express the uncertainty associated with a point estimate obtained from a sample.

Confidence Level

The confidence level, usually expressed as a percentage (for example, 95%), indicates how confident we are that the interval contains the true population parameter.

Eg : 95% confidence interval means that if you were to draw 100 different samples from the population and calculate the confidence interval for each, approximately 95 of those intervals would contain the true population parameter. (95% comes under the interval)

Point Estimate

A point estimate is a single value, calculated from a sample, that serves as the best guess or approximation for an unknown population parameter, such as the mean or standard deviation.

Eg : A sample of 50 students has an average (mean) height of **168 cm**.

- **Point estimate of population mean height = 168 cm**

Formula

$$\text{Confidence Interval} = \text{Point Estimate} \pm \text{Margin of Error}$$

Note:

Confidence Interval is created for parameters and not statistics. Statistics help us get the confidence interval for a parameter.

There are several ways to calculate a Confidence Interval (CI), But we will focus on 2

- 1. Confidence Interval for Population Mean (σ known)**
- 2. Confidence Interval for Population Mean (σ unknown)**

1. Confidence Interval for Population Mean (σ known)

Z-interval

Used when:

- Population standard deviation σ is known
- Sample size is large ($n \geq 30$) or population is normal
- Random Sampling → The data must be collected using a random sampling method

Confidence Interval Formula (σ Known)

$$CI = \bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

Where:

- $\bar{x}$ = sample mean
- σ = population standard deviation
- n = sample size
- $z_{\alpha/2}$ = critical Z-value (e.g., 1.96 for 95% confidence)

Factors Affecting Margin of Error

1. Confidence Level ($1 - \alpha$)
2. Sample Size
3. Population Standard Deviation

2. Confidence Interval for Population Mean (σ unknown)

t-interval

Used when:

- Population standard deviation is unknown

- Sample size is small
- Population is approximately normal
- Independent observations: The observations in the sample should be independent of each other
- Random Sampling → The data must be collected using a random sampling method

Confidence Interval Formula (t-interval)

$$CI = \bar{x} \pm t_{\alpha/2, n-1} \left(\frac{s}{\sqrt{n}} \right)$$

Where:

- $\bar{x}$ = sample mean
- s = sample standard deviation
- n = sample size
- $t_{\alpha/2, n-1}$ = critical t-value with $n - 1$ degrees of freedom

$n - 1$ determines the degrees of freedom of the t-distribution, which affects the critical t-value and hence the margin of error and width of the confidence interval.

Note:

- As degrees of freedom increase, the t-distribution approaches the normal distribution
- The t-distribution does not exist as a physical or natural phenomenon; it is a theoretical (mathematical) distribution used for statistical inference.
- The Z confidence interval is narrower than the t confidence interval because the t-distribution has heavier tails and uses a larger critical value to account for the additional uncertainty from estimating the population standard deviation.

For large samples ($n \geq 30$), the t-distribution closely approximates the z-distribution, so z may be used as an approximation.

Hypothesis Testing

A statistical hypothesis test is a method of statistical inference that uses sample data to determine whether there is enough evidence to reject a stated claim (null hypothesis) about a population parameter.

Null Hypothesis (H_0)

The null hypothesis is a statement that there is no effect, no difference, or no relationship between the variables in the population. It represents the status quo or default assumption and is assumed to be true unless there is strong statistical evidence against it.

Alternative Hypothesis (H_1 or H_a)

The alternative hypothesis contradicts the null hypothesis and states that a real effect, difference, or relationship exists in the population. It represents the research claim the investigator seeks evidence for.

Rule of thumb:

What you want to prove → Alternative hypothesis

What you assume by default → Null hypothesis

Note :

- Failing to reject H_0 does NOT mean H_0 is true “There is insufficient evidence to support the alternative hypothesis”
- Hypothesis testing evaluates whether sample evidence is strong enough to reject the null hypothesis in favor of the alternative hypothesis.

Hypothesis testing can be conducted using two equivalent methods:

1. The rejection region (critical value) approach
2. The p-value approach

Rejection region (critical value) approach

- Step 1: Formulate Null and Alternative Hypotheses
- Step 2: Select a Significance Level (α)
- Step 3: Check Assumptions
- Step 4: Decide Which Test is Appropriate
- Step 5: State the Relevant Test Statistic
- Step 6: Conduct the Test
- Step 7: Reject or Not Reject the Null Hypothesis
- Step 8: Interpret the Result

Question 1

Suppose a company is evaluating the impact of a new training program on the productivity of its employees. The company has data on the average productivity of its employees before implementing the training program. The average productivity was 50 units per day with a known population standard deviation of 5 units. After implementing the training program, the company measures the productivity of a random sample of 30 employees. The sample has an average productivity of 53 units per day. The company wants to know if the new training program has significantly increased productivity.

$\mu = 50, \sigma = 5, n = 30, \bar{x} = 53$

(i) $H_0: \mu = 50, H_a: \mu > 50$

(ii) $\alpha = 0.05$

(iii) Normality valid ($n \geq 30$)

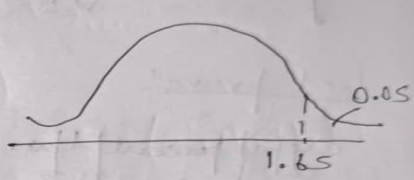
(iv) z-test = $\frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

$= \frac{53 - 50}{5 / \sqrt{30}} = 3.29$

(v) At a significance level $\alpha = 0.05$, the critical z-value for one-tailed test is 1.645

$= 1 - \alpha \rightarrow 1 - 0.05 \rightarrow 0.95$
and by using z-table (0.95) $\rightarrow 1.65$

(vi) Since, the calculated z-value (3.29) exceeds the critical value (1.645) we reject the null hypothesis.



Question 2

Suppose a snack food company claims that their Lays wafer packets contain an average weight of 50 grams per packet. To verify this claim, a consumer watchdog organization decides to test a random sample of Lays wafer packets. The organization wants to determine whether the actual average weight differs significantly from the claimed 50 grams. The organization collects a random sample of 40 Lays wafer packets and measures their weights. They find that the sample has an average weight of 49 grams, with a known population standard deviation of 4 grams.

① $H_0: \mu = 50$ $H_a: \mu \neq 50$

2) $\alpha = 0.05$

3) Normality.

4) z-test

$$z = \frac{49 - 50}{4/\sqrt{40}} = -1.58$$

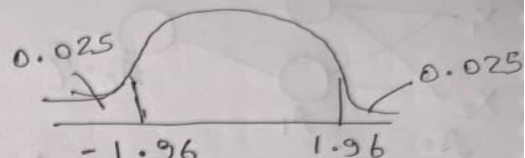
5) At a significance level of $\alpha = 0.05$ for a two tail test the critical z-values are ± 1.96 .
$$= 1 - 0.025$$

$$= 0.975$$

for 0.975 the z-values = 1.96

6) Since the calculated z-value = -1.58 falls between -1.96 and 1.96, it lies within the acceptance region.

$\therefore$ we fail to reject the null hypothesis.
There is insufficient evidence to conclude that the actual average weight of Lays wafer packets differ significantly from the claimed 50 grams.

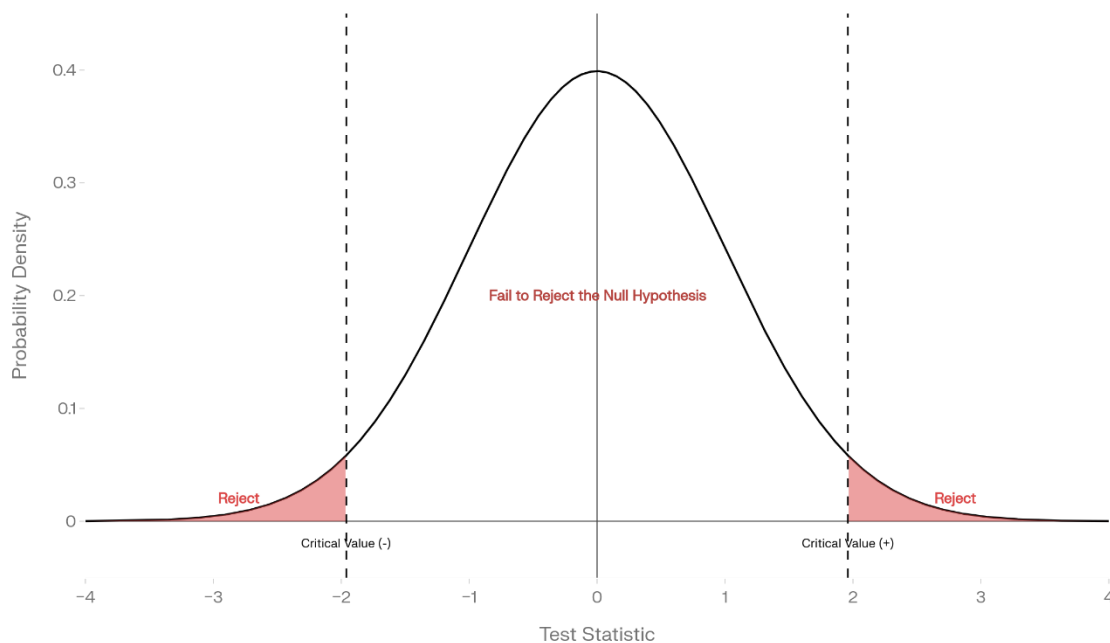


Significance Level (α)

The **significance level**, denoted as α (alpha), is a predetermined threshold used in hypothesis testing to determine whether the null hypothesis should be rejected or not. It represents the probability of rejecting the null hypothesis when it is actually true, also known as a **Type I error**.

Two-Tailed Hypothesis Test Critical Regions

Standard normal distribution with $\alpha = 0.05$



Problems with the Rejection Region Approach

- Does not show the strength of evidence
- Highly dependent on chosen significance level (α)

Note:

The rejection region approach provides only a binary decision and does not measure the strength of evidence against the null hypothesis, making interpretation less informative than the p-value approach.

P-Value approach

The p-value is the probability of obtaining a sample result as extreme as, or more extreme than, the observed sample—assuming that the null hypothesis (H_0) is true.

The p-value quantifies the strength of evidence against the null hypothesis by measuring how likely the observed data (or more extreme) would be if the null hypothesis were true.

Steps in the P-Value Approach

Step 1: State the Hypotheses

Step 2: Choose the Significance Level

Step 3: Compute the Test Statistic

Step 4: Calculate the p-Value

Step 5: Decision Rule

Step 6: Conclusion (in Context)

Question 1 - Suppose a company is evaluating the impact of a new training program on the productivity of its employees. The company has data on the average productivity of its employees before implementing the training program. The average productivity was 50 units per day. After implementing the training program, the company measures the productivity of a random sample of 30 employees. The sample has an average productivity of 53 units per day and the pop std is 4. The company wants to know if the new training program has significantly increased productivity.

① Given information
 $\mu = 50, \bar{x} = 53, \sigma = 4, n = 30, \alpha = 0.05$

② Hypothesis $\rightarrow$ right-tailed test
 $H_0: \mu = 50$ (no increase)
 $H_1: \mu > 50$ (increase)

③ Test Normality, $n = 30$

④ Test $Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{53 - 50}{4 / \sqrt{30}} = 4.11$

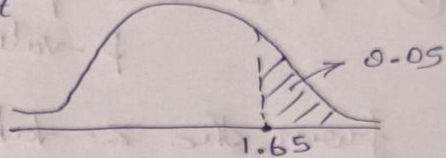
① Rejection Region Approach:

① critical value of right-tailed test at $\alpha = 0.05$
 $Z_{\alpha} = 1.645$
 $= 1.65$

② Decision Rule.
Reject H_0 if $Z > 1.65$

③ Compare
 $Z > 1.65$
 $4.11 > 1.65$

④ Reject H_0 (Null Hypothesis)



② P-value Approach.

① Find the p-value.

$$p\text{-value} = P(Z > 4.11)$$

↳ area above 4.11

from the z-table

$$P(Z > 4.11) \approx 0.00002$$

$Z > 4.11 = 99998$
$1 - 99998$
$= 0.00002$

② Compare with α

$$p\text{-value} = 0.00002 < 0.05$$

③ Decision

Reject the H_0 (Null Hypothesis)

④ How strong is the evidence

$$p\text{-value} = 0.00002$$

This means.

There is only a 0.002% chance of seeing such data if H_0 is true.

Question 2

Suppose a snack food company claims that their Lays wafer packets contain an average weight of 50 grams per packet. To verify this claim, a consumer watchdog organization decides to test a random sample of Lays wafer packets. The organization wants to determine whether the actual average weight differs significantly from the claimed 50 grams. The organization collects a random sample of 40 Lays wafer packets and measures their weights. They find that the sample has an average weight of 49 grams, with a pop standard deviation of 5 grams.

① Given information
 $\mu = 50$, $\bar{x} = 49$, $\sigma = 5$, $n = 40$, $\alpha = 0.05$

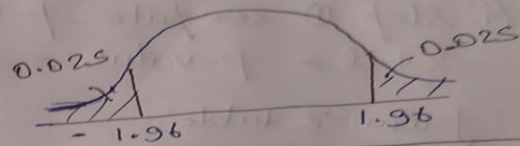
② Hypothesis.

$$H_0: \mu = 50$$
$$H_1: \mu \neq 50$$

③ Normality - valid $n = 40$

④ Test:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{49 - 50}{5 / \sqrt{40}} = -1.27$$



$$\alpha = 0.05$$
$$\text{two-tail} = 0.025$$
$$1 - 0.025 = 0.975$$
$$Z\text{-critical} = 1.96$$

⑤ Rejection Region Approach.

① Critical values.

for two-tailed test $\alpha = 0.05$

$$Z_{\alpha/2} = \pm 1.96$$

② Decision Rule:

$$Z < -1.96 \text{ or } Z > 1.96$$

③ Compare:

$$Z = -1.27$$

$$-1.96 < -1.27 < 1.96$$

→ Do not Reject H_0
fail to reject H_0

⑥ P-value Approach.

① Step 1 compute p-value.

$$\text{2 tail} \rightarrow p\text{-value} = 2P(Z < -1.27)$$

from Z table:

$$P(Z < -1.27) = 0.102$$

$$p\text{-value} = 2 \times 0.102 = 0.204$$

$$\text{Using Z table}$$
$$-1.27 \rightarrow 0.102$$

③ compare:

$$p\text{-value} = 0.204 > 0.05$$

Do not reject H_0 (Null Hypothesis)

④ How strong.

$$p = 0.204$$

$$= 20.4\%$$

that mean,

If the company's claim ($\mu = 50$) is true, there is about a 20% chance of getting a sample result like this just due to random sample.

T-test

A t-test is a statistical hypothesis test used to determine whether there is a significant difference between means. It is based on the t-distribution and is commonly used when:

- The population standard deviation is unknown, and
- The sample size is small (typically $n < 30$), though it can also be used for larger samples.

A t-test is a hypothesis test based on the t-distribution used to compare means when the population standard deviation is unknown.

Types of t-Tests

1. One-Sample t-Test
2. Independent Two-Sample t-Test
3. Paired t-Test (Dependent Two-Sample t-Test)

One-Sample t-Test

A one-sample t-test checks whether a sample mean differs from the population mean.

The one-sample t-test compares the mean of a single sample to a known or hypothesized population mean.

Assumptions of a One-Sample t-Test

1. **Normality**
2. **Independence**
3. **Random Sampling**
4. **Unknown Population Standard Deviation**

Question 1:

Suppose a manufacturer claims that the average weight of their new chocolate bars is 50 grams, we highly doubt that and want to check this so we drew out a sample of 25 chocolate bars and measured their weight, the sample mean came out to be 49.7 grams and the sample std deviation was 1.2 grams. Consider the significance level to be 0.05

Given
 $\mu = 50$, $n = 25$, $\bar{x} = 49.7$, $\sigma = 1.2$, $\alpha = 0.05$

(i) Hypothesis.
 $H_0: \mu = 50$
 $H_1: \mu \neq 50$

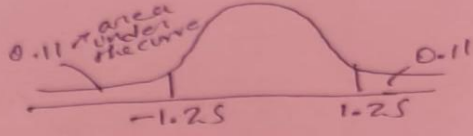
(ii) Test statistic
 $t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{49.7 - 50}{1.2/\sqrt{25}} = -1.25$

(iii) degree of freedom
 $df = n - 1$
 $= 24$

P-value Approach
 Compute P-value.
 Two-tailed test
 from t-table ($df = 24$)
 $p\text{-value} = 2P(T > |1.25|)$
 $P(T > 1.25) = 0.112$
 $p\text{-value} = 2 \times 0.112 = 0.224$

Code to find p-value.
 from scipy.stats import t
 $t.cdf(-1.25, 24)$
 $= 0.112$ (t value) $\rightarrow df$

Compare with α
 $0.224 > 0.05$
 * Do not reject H_0 (Null Hypothesis)



Code One Sample T-test

Independent Two-Sample t-Test

An independent two-sample t-test, also known as an unpaired t-test, is a statistical method used to compare the means of two independent groups to determine if there is a significant difference between them.

Assumptions for the test:

1. Independence of observations
2. Normality
3. Equal variances (Homoscedasticity)
4. Random sampling

Question 1:

Suppose a website owner claims that there is no difference in the average time spent on their website between desktop and mobile users. To test this claim, we collect data from 30 desktop users and 30 mobile users regarding the time spent on the website in minutes. The sample statistics are as follows:

desktop users = [12, 15, 18, 16, 20, 17, 14, 22, 19, 21, 23, 18, 25, 17, 16, 24, 20, 19, 22, 18, 15, 14, 23, 16, 12, 21, 19, 17, 20, 14]

mobile_users = [10, 12, 14, 13, 16, 15, 11, 17, 14, 16, 18, 14, 20, 15, 14, 19, 16, 15, 17, 14, 12, 11, 18, 15, 10, 16, 15, 13, 16, 11]

Code For Users Independent Two-Sample t-Test

The image shows a handwritten solution for an independent two-sample t-test. It includes the following steps:

- Sol:-** Normality $\rightarrow$ Shapiro, Equal variance $\rightarrow$ Levene.
- Hypotheses:**
 $H_0 = \mu_d = \mu_m$
 $H_a = \mu_d \neq \mu_m$
- t-statistic formula:**
$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$
- Calculation:**
$$t = \frac{18.5 - 14.3}{\sqrt{\frac{(3.5)^2}{30} + \frac{(2.7)^2}{30}}} = -5.25$$
- Degree of freedom:**
$$\begin{aligned} \text{degree of freedom} &= n_1 + n_2 - 2 \\ &= 30 + 30 - 2 = 58 \end{aligned}$$
- Using code:**
$$t.cdf(-5.25, 58)$$

$$p\text{-value} = 2.256e^{-06}$$

$$= 2.256e^{-06}$$
- Conclusion:**
$$[2.256e^{-06} < 0.05] \text{ Reject } H_0$$

There is also a small hand-drawn graph of a normal distribution curve with the area to the left of -5.25 shaded, representing the p-value.

Question 2 Code Solution Independent Two-Sample T-Test question 2

Paired t-Test (Dependent Two-Sample t-Test)

A paired two-sample t-test is used to compare the means of two related samples by testing whether the mean of their paired differences is significantly different from zero.

A paired two-sample t-test, also known as a dependent t-test or paired-samples t-test, is used to determine whether there is a significant difference between the means of two related or dependent samples.

Common Scenarios for Using a Paired t-Test

1. Before-and-after studies
2. Matched or correlated samples Eg (Siblings , Twins)

Assumptions of the Paired t-Test

1. Paired Observations
2. Normality of Differences
 - Normality of differences means that the distribution of the paired differences (Sample2 – Sample1) is approximately normal.
3. Independence of Pairs

Question

Let's assume that a fitness center is evaluating the effectiveness of a new 8-week weight loss program. They enroll 15 participants in the program and measure their weights before and after the program. The goal is to test whether the new weight loss program leads to a significant reduction in the participants' weight.

Solution Paired t-Test (Dependent Two-Sample t-Test)

Before the program:

[80, 92, 75, 68, 85, 78, 73, 90, 70, 88, 76, 84, 82, 77, 91]

After the program:

[78, 93, 81, 67, 88, 76, 74, 91, 69, 88, 77, 81, 80, 79, 88]

$$t = \frac{\bar{x}_{diff}}{s_{diff} / \sqrt{n}} = 0.1048$$
$$p\text{ value} = \text{stats.t.cdf}(0.1048, df)$$
$$p\text{ value} = 0.45899 < 0.05$$

So, Reject H_0

Type I Error (False Positive)

A Type I error occurs when the null hypothesis (H_0) is rejected even though it is true.

A Type I error occurs when we reject the null hypothesis even though it is true.

Example:

A person is found guilty even though they are actually innocent

Type II Error (False Negative)

A Type II error occurs when the null hypothesis is not rejected even though it is false.

A Type II error occurs when we fail to reject the null hypothesis even though it is false.

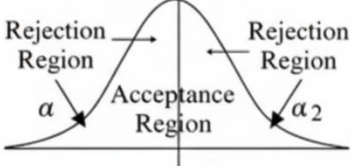
Example :

A person is actually guilty, but the court finds them not guilty

Trade-off Between Type I and Type II Errors

Reducing the probability of a Type I error increases the probability of a Type II error, and balancing this trade-off is achieved by selecting appropriate values of α and β .

One-Sided vs Two-Sides Tailed Test

One-Tailed Test (Left Tail)	Two-Tailed Test	One-Tailed Test (Right Tail)
$H_0 : \mu_x = \mu_0$ $H_1 : \mu_x < \mu_0$	$H_0 : \mu_x = \mu_0$ $H_1 : \mu_x \neq \mu_0$	$H_0 : \mu_x = \mu_0$ $H_1 : \mu_x > \mu_0$
		

One-Sided (One-Tailed) Test

A **one-sided (one-tailed) test** is used when the researcher is interested in detecting an effect in **only one specific direction**—either **greater than** or **less than** the value stated in the null hypothesis.

Two-Sided (Two-Tailed) Test

A **two-sided (two-tailed) test** is used when the researcher wants to detect **any difference**, regardless of direction—whether the value is **greater or smaller** than that stated in the null hypothesis.

Note:

A one-tailed test examines an effect in a single direction, while a two-tailed test examines differences in both directions, with the significance level divided accordingly.

Where Can Hypothesis Testing Be Applied?

Hypothesis testing is used to **check whether observed differences, effects, or relationships in data are real or just due to random chance**, such as comparing groups, evaluating treatments, testing relationships, checking data fit, testing independence, and running A/B experiments.

Hypothesis Testing in Machine Learning Applications

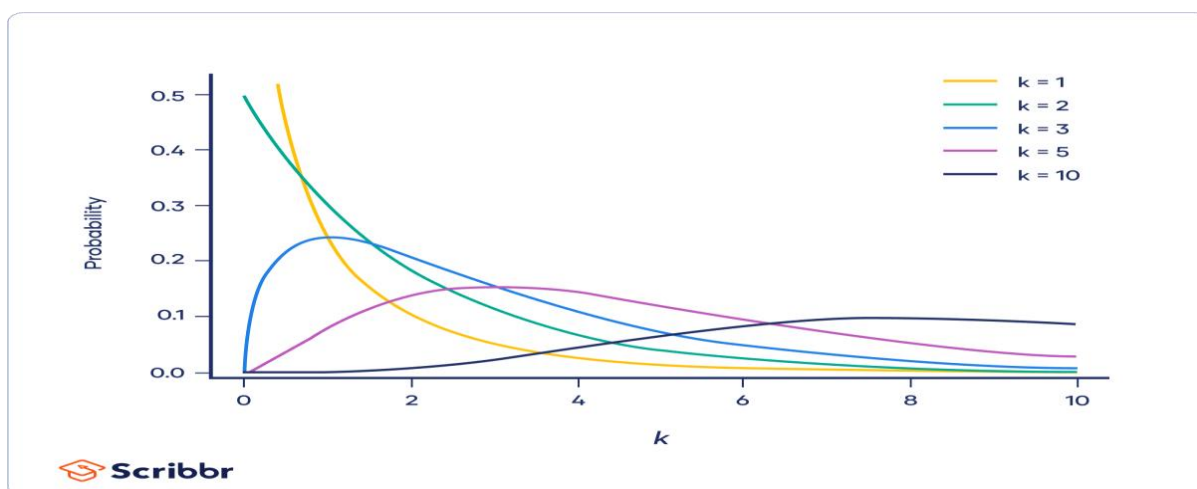
In machine learning, hypothesis testing is used to compare models, select important features, validate hyperparameter improvements, and check model assumptions, ensuring performance gains are statistically meaningful and not due to random chance.

Chi-Square distribution

If you take a sample from a *standard normal distribution* (mean = 0, variance = 1) and square it, the resulting random variable follows a Chi-Square distribution with 1 degree of freedom.

Chi-Square distribution arises from the sum of squares of independent standard normal variables.

The Chi-Square (χ^2) distribution is a continuous probability distribution used mainly in statistical hypothesis testing, especially when working with categorical data.



[Chi-Square distribution.ipynb](#)

Some key properties of the Chi-Square distribution are:

- It is a continuous distribution, defined for non-negative values.
- It is positively skewed, with the degree of skewness decreasing as the degrees of freedom increase.
- The mean of the Chi-Square distribution is equal to its degrees of freedom, and its variance is equal to twice the degrees of freedom.
- As the degrees of freedom increase, the Chi-Square distribution approaches the normal distribution in shape.

Applications of Chi-Square (χ^2) Distribution

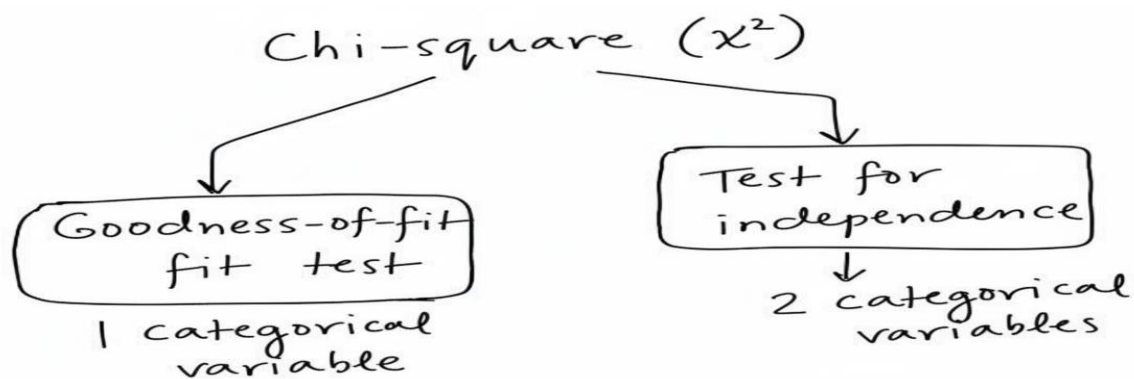
1. Chi-Square Goodness-of-Fit Test
2. Chi-Square Test for Independence
3. Chi-Square Test for Homogeneity

Chi-Square (χ^2) Test

The **Chi-Square test** is a **non-parametric statistical hypothesis test** used to analyze **categorical data**. It helps determine whether:

- An **observed frequency distribution** differs significantly from an **expected distribution**, or
- There is a **significant association** between two **categorical variables**.

The test statistic follows a **Chi-Square (χ^2) distribution**.



Main Types of Chi-Square Tests

1 Chi-Square Goodness-of-Fit Test

- **Purpose:**
To check whether the observed frequencies of a **single categorical variable** match an **expected theoretical distribution**.
- **Compares:**
Observed frequencies vs Expected frequencies

Example:

Checking whether a die is fair (each face expected with probability 1/6).

2 Chi-Square Test for Independence (Association)

- **Purpose:**
To determine whether **two categorical variables** are **independent** or **associated**.

- **Compares:**
Observed cell counts vs Expected cell counts (under independence assumption)

Example:

Testing whether **gender** and **product preference** are related.

1 . Chi-Square Goodness-of-Fit Test

The Chi-Square Goodness-of-Fit test is a statistical hypothesis test used to determine whether the observed frequency distribution of a single categorical variable is consistent with an expected theoretical distribution.

This test evaluates whether sample data follow a specified probability distribution—such as the uniform, binomial, or Poisson distribution—by comparing observed frequencies with expected frequencies derived from the assumed model.

Steps for Chi-Square Goodness-of-Fit Test

1. State the Hypotheses

- Null Hypothesis (H_0):

The observed data follow the expected theoretical distribution.

- Alternative Hypothesis (H_1):

The observed data do not follow the expected theoretical distribution.

2. Calculate Expected Frequencies

For each category, compute the expected frequency based on the theoretical distribution and total sample

3. Compute the Chi-Square Test Statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where

- O_i = observed frequency in category i
- E_i = expected frequency in category i
- The summation is over all categories

4. Determine Degrees of Freedom

$$df = k - 1$$

where

- k = number of categories

5. Find the p-Value

Using the calculated χ^2 value and degrees of freedom, obtain the **p-value** from the chi-square distribution table or statistical software.

6. Make the Decision

- If **p-value** $\leq \alpha$ (significance level, e.g., 0.05), **reject H_0**
- If **p-value** $> \alpha$, **fail to reject H_0**

In a Chi-Square Goodness-of-Fit test, you have two equivalent decision methods:

1 Critical Value Method

- Compare calculated χ^2 with the table value
- If $\chi^2 > \chi^2_{\text{critical}} \rightarrow \text{Reject } H_0$
- If $\chi^2 \leq \chi^2_{\text{critical}} \rightarrow \text{Fail to reject } H_0$

✓ Used often in exams

2 p-Value Method

- Calculate p-value using χ^2 and degrees of freedom
- If p-value $\leq \alpha \rightarrow \text{Reject } H_0$
- If p-value $> \alpha \rightarrow \text{Fail to reject } H_0$

✓ Common in software & research

Assumptions of Chi-Square Goodness-of-Fit :

- Observations must be **independent**.
- Data must be **categorical** with mutually exclusive categories.
- **Expected frequency ≥ 5** in each category.
- The **theoretical distribution is fixed in advance** (not chosen after seeing data).

Note:

The **Chi-Square Goodness-of-Fit test** is a **non-parametric test** because it does not require assumptions about population parameters such as the mean or standard deviation, nor does it assume a specific underlying data distribution. Instead, it compares **observed categorical frequencies** with **expected frequencies** from a hypothesized model, focusing only on counts rather than parameter estimates.

Example 1:

Suppose we have a six-sided fair die, and we want to test if the die is indeed fair. We roll the die 60 times and record the number of times each side comes up. We will use the Chi-Square Goodness-of-Fit test to determine if the observed frequencies are consistent with a fair die (that is, a uniform distribution of the sides).

Observed frequencies:

Side 1: 12 times Side 2: 8 times Side 3: 11 times Side 4: 9 times Side 5: 10 times

Side 6: 10 times

Given: $n = 60$, $k = 6$, fair die $= \frac{1}{6}$

① Hypotheses
 H_0 : The die is fair (uniform distribution)
 H_1 : The die is not fair.

② Expected frequencies (E) $E = n/k$
 $E = \frac{60}{6} = 10$ for each

③ Compute chi square statistic

Observed	12	8	11	9	10	10
Expected	10	10	10	10	10	10
$\frac{(O - E)^2}{E}$	0.40	0.40	0.10	0.10	0.00	0.00

$\chi^2 = \sum \frac{(O - E)^2}{E}$
 $\chi^2 = 1.0$

④ Degree of freedom
 $df = k - 1$
 $= 6 - 1$
 $= 5$

(5) Decision

(i) By using Critical value Method.

- * Critical value at $\alpha = 0.05$ & $df = 5$
 $= 11.07$ (using chi-table)
- * Calculated $\chi^2 = 1.00$

$1.00 < 11.07$ Fail to reject H_0

(2) By using p-value.

Using Software.

from scipy.stats import chi2
 $p\text{-value} = 1 - \text{chi2.cdf}(1.00, df=5)$

$p\text{-value} = 0.962$

$0.05 < 0.962$

* fail to Reject H_0 (Null Hypotheses)

Example 2:

Suppose a marketing team at a retail company wants to understand the distribution of visits to their website by day of the week. They have a hypothesis that visits are uniformly distributed across all days of the week, meaning they expect an equal number of visits on each day. They collect data on website visits for four weeks and want to test whether the observed distribution matches the expected uniform distribution.

Observed frequencies (number of website visits per day of the week for four weeks):

Monday: 420

Tuesday: 380

Wednesday: 410

Thursday: 400

Friday: 410

Saturday: 430

Sunday: 390

Given:

$$k = 7$$

$$\text{Total visits} = 420 + 380 + 410 + 400 + 410 + 430 + 390$$

$$n = 2840$$

Step 1 Hypothesis

H_0 : Visits are uniformly distributed across all days.

H_1 : Visits are not uniformly distributed

Step 2 Expected frequencies.

$$E = n/k = \frac{2840}{7} = 405.71 \approx 406$$

Step 3 Compute chi-square statistic.

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

	Observed	Expected E	$\frac{(O - E)^2}{E}$
Monday	420	405.71	$\frac{(420 - 405.71)^2}{405.71} = 0.52$
Tuesday	380	405.71	$\frac{(380 - 405.71)^2}{405.71} = 1.66$
Wednesday	410	405.71	0.05
Thursday	400	405.71	0.08
Friday	410	405.71	0.05
Saturday	430	405.71	1.48
Sunday	390	405.71	0.63

$$\chi^2 = 0.52 + 1.66 + 0.05 + 0.08 + 0.05 + 1.48 + 0.63 \Rightarrow 4.47$$

Step 4: Degree of freedom

$$df = k - 1 \Rightarrow 7 - 1 \Rightarrow 6$$

$$df = 6$$

Step 5 Decision

① By using critical value.

from chi-square table, $\chi_{0.05, 6}^2 = 12.59$

$$4.47 < 12.59$$

Fail to reject $H_0 \rightarrow$ No significant evidence against uniform distribution.

② By using p-value.

By software p-value = 0.61

$$0.05 < 0.61$$

p-value > 0.05 , so fail to reject H_0

Example 3:

A survey of 800 families in a village with 4 children each revealed the following distribution:

Number of boys in a family:	0,	1,	2,	3,	4
Number of families:	32,	178,	290,	236,	64

Is this data consistent with the result that male and female births are equally probable?

Given:
Total families $n = 800$
Number of children per family: 4

Step 1 Hypothesis
 H_0 : Distribution of boys per family follows a binomial distribution with $p = 0.5$
 H_1 : Distribution of boys per family does not follow a binomial distribution with $p = 0.5$

Step 2 Compute Expected frequencies.
$$P(X=k) = {}^n C_k p^k (1-p)^{n-k} \quad n=4$$
$$= {}^n C_k p^k (1-p)^{n-k} \quad p = \frac{1}{2}$$

Probabilities:
* $P(0 \text{ boys}) = {}^4 C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = \frac{1}{16} \times 800 = 50$
* $P(1 \text{ boys}) = {}^4 C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = 0.25 \times 800 = 200$
* $P(2 \text{ boys}) = {}^4 C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = 0.375 \times 800 = 300$
* $P(3 \text{ boys}) = {}^4 C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 = 0.25 \times 800 = 200$
* $P(4 \text{ boys}) = {}^4 C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = 0.0625 \times 800 = 50$

Step 3 Compute chi-square statistic

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Number of boys			$(O - E)^2 / E$
0	32	50	$(32 - 50)^2 / 50 = 6.48$
1	178	200	$(178 - 200)^2 / 200 = 2.42$
2	290	300	$\frac{(290 - 300)^2}{300} = 0.33$
3	236	200	$\frac{(236 - 200)^2}{200} = 6.48$
4	64	50	$\frac{(64 - 50)^2}{50} = 3.92$

$$\chi^2 = 6.48 + 2.42 + 0.33 + 6.48 + 3.92 \Rightarrow \boxed{19.63}$$

Step 4 Degree of freedom.

$$df = k - 1 \Rightarrow 5 - 1 = 4$$

Step 5 Critical value at $\alpha = 0.05$

From chi-square table: $\chi^2_{0.05, 4} \approx 9.49$

Step 6 Decision

$\boxed{19.63 > 9.49}$

Reject $H_0 \rightarrow$ The observed data do not fit the expected binomial distribution

Chi-Square Goodness-of-Fit Test.ipynb

2. Chi-Square test for independence

The Chi-Square test for independence, also known as the Chi-Square test for association, is a statistical test used to determine whether there is a significant association between two categorical variables in a sample. It helps to identify whether the occurrence of one variable is dependent on the occurrence of the other variable, or if they are independent of each other.

The test is based on comparing the observed frequencies in a contingency table (a table that displays the frequency distribution of the variables) with the frequencies that would be expected under the assumption of independence between the two variables.

Steps

1. State the hypotheses
2. Create a contingency table
3. Calculate expected frequencies
4. Compute the Chi-Square test statistic

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed frequency in each cell and E_{ij} is the expected frequency.

5. Determine the degrees of freedom
6. Obtain the critical value or p-value
7. Make a decision

Assumptions of Chi-Square Test for Independence :

- Observations must be independent.
- Both variables must be categorical (nominal or ordinal).
- Expected frequency ≥ 5 in each cell (large enough sample size).
- Marginal totals are fixed before data collection.

Example 1:

A researcher wants to investigate if there is an association between the level of education (categorical variable) and the preference for a particular type of exercise (categorical variable) among a group of 150 individuals. The researcher collects data and creates the following contingency table

Education	Yoga	Running	Swimming	Total
High School	15	20	10	45
Bachelor's	20	30	15	65

Education	Yoga	Running	Swimming	Total
Master's or PhD	5	15	20	40
Total	40	65	45	150

Step 1: State the Hypotheses

- H_0 (Null Hypothesis):
Level of education and exercise preference are **independent**.
- H_1 (Alternative Hypothesis):
Level of education and exercise preference are **associated**.

Step 2: Compute Expected Frequencies

Formula:

$$E_{ij} = \frac{(\text{Row total})(\text{Column total})}{\text{Grand total}}$$

Education	Yoga	Running	Swimming
High School	12	19.5	13.5
Bachelor's	17.33	28.17	19.5
Master's / PhD	10.67	17.33	12

(All expected frequencies $\geq 5 \rightarrow$ assumption satisfied)

Step 3: Compute Chi-Square Statistic

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Education × Exercise	$(O - E)^2 / E$
High School–Yoga	0.75

Education × Exercise	$(O-E)^2 / E$
High School–Running	0.01
High School–Swimming	0.91
Bachelor–Yoga	0.41
Bachelor–Running	0.12
Bachelor–Swimming	1.04
Master–Yoga	3.01
Master–Running	0.31
Master–Swimming	5.33

$$\chi^2 \approx 11.89$$

Step 4: Degrees of Freedom

$$\begin{aligned}
 Df &= (r-1)(c-1) \\
 &= (3-1)(3-1) \\
 &= 4
 \end{aligned}$$

Step 5: Critical Value ($\alpha = 0.05$)

From chi-square table:

$$\chi_{0.05, 42} = 9.49$$

Step 6: Decision

$$\chi_{\text{calc}2} = 11.89 > 9.49$$

Reject H_0

[Chi-Square test for independence .ipynb](#)

Note:

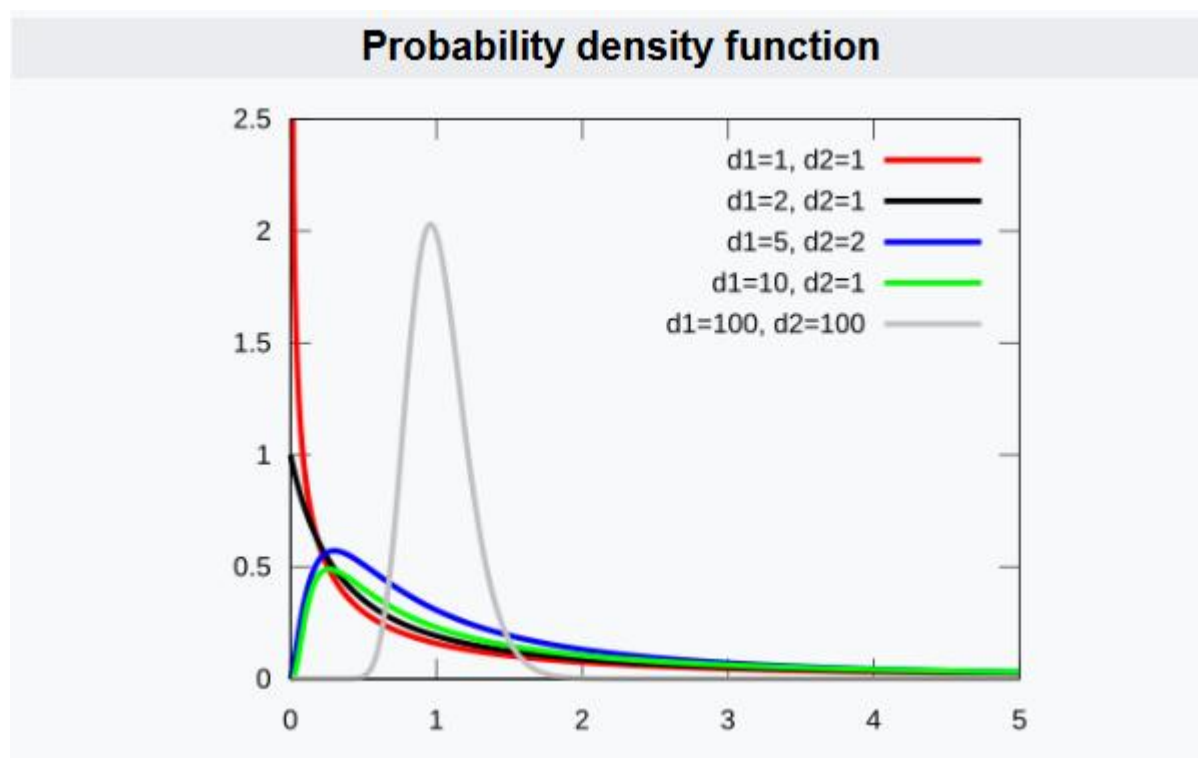
In machine learning, the Chi-Square test is mainly used for feature selection, categorical data analysis, model evaluation, and decision tree construction.

F-Distribution

The F-distribution is a continuous probability distribution used in statistical hypothesis testing and analysis of variance (ANOVA).

The F-distribution is a right-skewed continuous distribution used to compare variances and test group mean differences in ANOVA and regression models.

It is also known as the Fisher-Snedecor distribution, named after Ronald Fisher and George Snedecor, two prominent statisticians.



Degrees of Freedom

- The F-distribution is defined by two parameters - the degrees of freedom for the numerator (df1) and the degrees of freedom for the denominator (df2). It determine the shape of the distribution

Positively Skewed and Bounded

- The F-distribution is **positively (right) skewed**
- It is **bounded at zero** ($F \geq 0$)
- As **df₁ and df₂ increase**, the distribution becomes **less skewed** and more symmetric (may be Normal distribution)

F-Statistic

The F-statistic is computed as:

$$F = \frac{\text{Variance}_1}{\text{Variance}_2} \quad \text{or} \quad F = \frac{\text{Mean Square Between}}{\text{Mean Square Within}}$$

- The statistic follows an **F-distribution** under the null hypothesis
- The calculated F value is compared with a **critical value** from the F-table to determine significance

Note :

The F-distribution is formed by dividing two independent Chi-square distributions, each scaled by its respective degrees of freedom.

Testing Equality of Variances

- Used to test whether **two populations have equal variances**
- Common in quality control and experimental analysis

ANOVA (Analysis of Variance)

One-way ANOVA (Analysis of Variance) is a statistical method used to compare the means of three or more independent groups to determine if there are any significant differences between them.

It is an **extension of the independent two-sample t-test**, which is limited to comparing only two group means.

The term “**one-way**” indicates that there is **only one independent variable (factor)** with **multiple levels (groups)**.

Core Idea (Intuition)

One-way ANOVA compares:

- Variation between group means
vs
- Variation within groups

If between-group variation is large relative to within-group variation, the group means are unlikely to be equal.

Steps in One-Way ANOVA

1. Define Hypotheses

2. Compute Means

- Calculate the **mean of each group**
- Calculate the **grand mean** (overall mean of all observations)

3. Calculate Sum of Squares (SS)

- **Between-Group Sum of Squares (SSB):**
Measures variation due to differences between group means
- **Within-Group Sum of Squares (SSW):**
Measures variation within each group (random/error variation)

4. Degrees of Freedom

- **Between groups:**

$$df_{between} = k - 1$$

- **Within groups:**

$$df_{within} = N - k$$

where

k = number of groups,

N = total number of observations

5. Mean Squares (MS)

- **Mean Square Between:**

$$MSB = \frac{SSB}{df_{between}}$$

- **Mean Square Within:**

$$MSW = \frac{SSW}{df_{within}}$$

6. F-Statistic

$$F = \frac{MSB}{MSW}$$

- The F-statistic follows an F-distribution
- Compare the calculated F value with the critical F value (or p-value)

Decision Rule

- If **F is large** or **p-value < α** , reject H_0
- Otherwise, fail to reject H_0

Example 1:

Height of 3 Group

A (group A)	B (group B)	C (group C)
3	1	8
6	8	6
3	9	10

Given $n=9, k=3$

① Hypotheses.
 $H_0: \mu_A = \mu_B = \mu_C$
 H_1 : atleast one of them is significant diffⁿ

② Compute Means.
 * mean of each group
 $\bar{x}_A = 4$
 $\bar{x}_B = 6$
 $\bar{x}_C = 9$

A	B	C
3	1	8
6	8	6
3	9	10
$\bar{x}_A = 4$	$\bar{x}_B = 6$	$\bar{x}_C = 9$

* Grand mean
 $\bar{x} = \frac{3+6+3+1+8+9+8+6+10}{9} = 6$

③ Calculate Sum of Squares.
 * SST $\rightarrow$ Sum of Square total
 $\sum (x_{ij} - \bar{x})^2 = (6-3)^2 + (6-6)^2 + (6-3)^2 + (6-1)^2 + (6-8)^2 + (6-9)^2 + (6-8)^2 + (6-9)^2 + (6-10)^2 \Rightarrow 76$

* SSW $\rightarrow$ Sum of Square Within-Group
 $\sum (\bar{x}_{A\bar{x}} - x_{Aj})^2 = (4-3)^2 + (4-6)^2 + (4-3)^2 + (6-1)^2 + (6-8)^2 + (6-9)^2 + (8-8)^2 + (8-6)^2 + (8-10)^2 \Rightarrow 57$

* $SSB \rightarrow$ Sum of Square Between Group.

$$SSB \rightarrow n_1(\bar{x}_1 - \bar{x})^2 + \dots + n_k(\bar{x}_k - \bar{x})^2$$

$3 \times (6-4)^2 + 3 \times (6-6)^2 + 3 \times (6-8)^2$
 grand mean group mean of A no. of sample of group C
 no. of sample of group A

$[SSB = 24]$

④ Degree of freedom.

* Within group = $N - k$
 $df_{within} = 9 - 3$
 $[df_{within} = 6]$

* Between group
 $df_{between} = k - 1$
 $[df_{between} = 2]$

⑤ $F = \frac{MSB}{MSW} = \frac{\frac{SSB}{df_{between}}}{\frac{SSW}{df_{within}}} = \frac{\frac{24}{2}}{\frac{50.2}{6}} = 1.4$

⑥ P-value using software.
 Stats. f. st (1.4, 2, 6) $\Rightarrow [0.31 > 0.05]$
 $[u_1 = u_2 = u_3]$ fail to reject the Null

[one_way_anova.ipynb](#)

Post-hoc Tests

Post-hoc tests (also called **multiple comparison tests** or **pairwise comparison tests**) are performed **after an ANOVA** when the **overall F-test is significant**.

Post-hoc tests are used after a significant ANOVA to identify which specific group means differ while controlling the family-wise error rate.

They are used to determine **which specific group means differ significantly** from each other.

- ANOVA tells *that* a difference exists
 Post-hoc tests tell *where* the difference exists

Why Post-hoc Tests:

- ANOVA tells us *whether* at least one group mean is different.
- Post-hoc tests tell us *which* specific groups are different from each other.

ANOVA cannot identify the exact pairs of groups that differ — it only gives an **overall yes/no decision**.

Common Post-hoc Tests

1. Bonferroni Correction

This method adjusts the significance level (α) by dividing it by the number of comparisons being made.

It is a conservative method that can be applied when making multiple comparisons, but it may have lower statistical power when a large number of comparisons are involved.

```
import scipy.stats as stats

for class1,class2 in [(1,2),(2,3),(3,1)]:
    print(class1,class2)
    print(stats.ttest_ind(df[df['Pclass'] == class1]['Age'], df[df['Pclass'] == class2]['Age']))
```



```
1 2
Ttest_indResult(statistic=5.485187676773201, pvalue=7.835568991415144e-08)
2 3
Ttest_indResult(statistic=3.927800191020872, pvalue=9.715078600777852e-05)
3 1
Ttest_indResult(statistic=-10.849122601201033, pvalue=6.134470007830625e-25)
```

Why Bonferroni correction is needed

A single t-test controls the **Type-I error rate (false positive)** at $\alpha = 0.05$.

But when you perform **multiple tests**, the probability of getting **at least one false positive increases**.

Here you performed:

$$m = 3 \text{ comparisons}$$

Without correction, overall (family-wise) error rate > 0.05 (3 t – test , so error =0.15)

So , we

With $\alpha = 0.05$ and $m = 3$ (no of test):

$$\alpha_{\text{Bonferroni}} = \frac{0.05}{3} \approx 0.0167$$

Each p-value must be < 0.0167 to be considered significant

2. Tukey's HSD (Honestly Significant Difference)

- Compares **all possible pairs of group means**
- Controls **FWER**
- Assumes:
 - **Equal variances**
 - Usually **equal or similar sample sizes**

```

from statsmodels.stats.multicomp import pairwise_tukeyhsd
import matplotlib.pyplot as plt

tukey = pairwise_tukeyhsd(endog=df['Age'], groups=df['Pclass'], alpha=0.05)

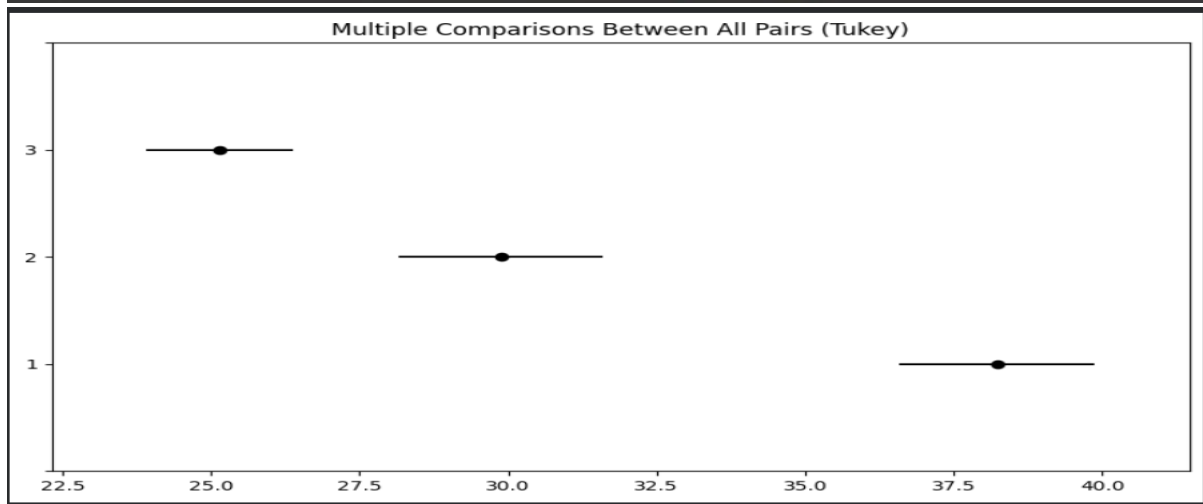
tukey.plot_simultaneous()

plt.vlines(x=49.57, ymin=-0.5, ymax=4.5, color="red")

pd.DataFrame(tukey.summary())

```

	0	1	2	3	4	5	6
0	group1	group2	meandiff	p-adj	lower	upper	reject
1	1	2	-8.3558	0.0	-11.7041	-5.0075	True
2	1	3	-13.0928	0.0	-15.9622	-10.2234	True
3	2	3	-4.737	0.0005	-7.6763	-1.7977	True



Note: Why t-test is not used for more than 2 (or 3+) groups

The t-test is not used for more than two groups because multiple t-tests inflate Type I error and make interpretation inefficient; ANOVA provides a single, controlled comparison.

Post-hoc Tests.ipynb

When to Use What (Titanic Context)

Situation	Test
Gender only (2 groups)	t-test
Pclass only (3 groups) or more	One-way ANOVA
Gender + Pclass together	Two-way ANOVA

F-Test

The F-test is a statistical test that checks whether groups or models are significantly different by comparing their variances.

It is used to answer questions like:

- Do the X variables explain Y? (Regression)
- Are two or more groups different? (ANOVA)
- Are two variances equal? (Variance comparison)

In all cases, the F-test uses a value called the F-statistic.

Breakdown in Simple Steps

F-Test (whole process):

1. Set null hypothesis
2. Set alternative hypothesis
3. Fit regression model
4. Calculate F-statistic (this is ONE step)
5. Compare with distribution
6. Find p-value
7. Make conclusion (significant or not)

Part	Meaning
F-Statistic	Your score in the exam (e.g., 12.4)
F-Test	The rule that decides whether you pass or fail
Prob(F-Statistic)	The chance that your score is high enough to pass (p-value) chance your score is real

T-Test

A **t-test** is a statistical test used to check whether the **difference between two values (or groups)** is **real or just due to random chance**.

Use of t-test in Machine Learning

1. To test significance of individual features (coefficients) in linear regression

While the **F-test** checks the **overall model**, the **t-test** checks **each variable separately**.

2. In Linear Regression

To check if an **individual feature ($X_1, X_2, \dots$)** actually affects the target (Y).

Performing a t-test for a simple linear regression, including the intercept term and using the p-value approach, involves the following steps:

1. State the null and alternative hypotheses for the slope and intercept coefficients:

- For the slope coefficient β_1
Null hypothesis (H_0): $\beta_1 = 0$
Alternative hypothesis (H_1): $\beta_1 \neq 0$
- For the intercept coefficient β_0
Null hypothesis (H_0): $\beta_0 = 0$
Alternative hypothesis (H_1): $\beta_0 \neq 0$

2. Estimate the slope and intercept coefficients (b_0 and b_1):

Using the sample data, calculate the slope (b_1) and intercept (b_0) coefficients for the regression model.

3. Calculate the standard errors for the slope and intercept coefficients ($SE(b_0)$ and $SE(b_1)$):

$$SE(b_1) = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{(n - 2) \sum (x_i - \bar{x})^2}}$$

$$SE(b_0) = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

4. Compute the t-statistics for the slope and intercept coefficients:

$$t_{b_1} = \frac{b_1 - \beta_1}{SE(b_1)}$$

$$t_{b_0} = \frac{b_0 - \beta_0}{SE(b_0)}$$

Feature	t-Test	F-Test
What it tests	Tests one coefficient at a time	Tests all coefficients together
Purpose	Check if a single feature affects Y	Check if model as a whole explains Y
Null Hypothesis	$\beta_i = 0 \rightarrow$ feature is useless	All β 's = 0 $\rightarrow$ model is useless
Alternative H_1	$\beta_i \neq 0 \rightarrow$ feature important	At least one $\beta \neq 0 \rightarrow$ model significant
Used for	Individual feature significance	Overall model significance
Where used	Linear regression, A/B test	Regression, ANOVA
Output	t-statistic + p-value	F-statistic + p-value
Interpretation	"Is this one variable useful?"	"Is the whole model useful?"
If test fails	Remove/ignore that feature	Model does NOT explain Y
Data comparison	Compares 2 groups	Compares ≥ 2 groups

